

Magic number 3 card trick

Here's another magic trick you can perform for a friend. You'll need a deck of cards and a calculator.

1. Ask your friend to secretly choose a card from the pack. It has to be a number card, not a face card like a jack, king, or queen. Tell your friend to remember their card, and place it face-down on the table.

2. Now secretly choose your own card, and place this face-down on the table too. For the trick to work, you have to choose a 3.

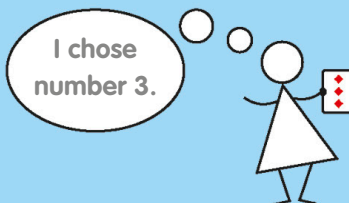
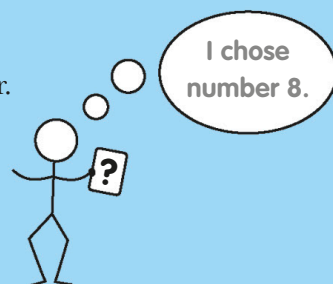
3. Once your friend has placed their card face-down on the table, give them the calculator. Tell them to enter the number they chose, and do this calculation:

Don't forget to press the = button after each one!



Multiply it by 2
Add 2
Multiply by 5
Subtract 7

4. Turn over their card, then your card. The numbers on the two cards will match the numbers on the calculator screen!



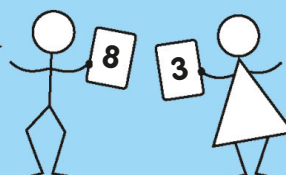
$$8 \times 2 = 16$$

$$+ 2 = 18$$

$$\times 5 = 90$$

$$- 7 = 83$$

Our numbers match the answer!



This technique also works with bigger numbers. If you add together the digits of the answer and get a two- or three-digit number, then you will need to add those digits together as well.

With 267 $\rightarrow 2 + 6 + 7 = 15$
and $1 + 5 = 6$
So 267 is a multiple of 3.

With 846 $\rightarrow 8 + 4 + 6 = 18$
and $1 + 8 = 9$
So 846 is a multiple of 3.

